16.07 Dynamics — Quiz 3: Bifilar Pendulum

Name: _____

The questions below refer to the T-shaped body and bifilar pendulum from Problem Set 3. Recall that for small oscillations, the period of the bifilar pendulum is

$$P = \frac{2\pi}{d}\sqrt{\frac{J\ell}{mg}},$$

where m is the mass of the body, J is its moment of inertia about the vertical axis through its center of mass, ℓ is the string length, and $2d$ is the string separation.

1. Suppose we measure the moment of inertia of the T-shaped body about a vertical axis that does *not* pass through its center of mass. Will this moment of inertia be smaller than, equal to, or larger than the value about the vertical axis through the center of mass? *Hint:* think about the parallel-axis theorem.

2. Suppose we make the T-shaped body from a material with twice the density (so that it's mass doubles), keeping everything else the same. Does its moment of inertia J change? If so, by what factor? Does the period P of the bifilar pendulum change?

3. Now suppose we double every dimension of the T-shaped body, but adjust the material density so that its mass m stays the same (and keep everything else the same). By what factor does J change? By what factor does the period P change?